IAC Provisioning Finance System

Course Name: DevOps Foundation

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year: 2025-2026

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1. Problem Statement & Objectives

1.1 Problem Statement

In modern financial systems, managing infrastructure manually is a complex and time-consuming process. Traditional infrastructure setup requires manual configuration of servers, networks, and software environments. This approach often leads to **inconsistencies, configuration errors, and increased deployment time**.

Financial applications require **high availability, security, and scalability**, which are difficult to maintain using manual infrastructure management. Additionally, repeated configuration across different environments such as development, testing, and production increases the chances of mistakes.

To overcome these challenges, there is a need for an automated infrastructure provisioning system that can create and manage resources efficiently. **Infrastructure as Code (IaC)** provides a solution by allowing infrastructure to be defined and managed through code, ensuring consistency and automation.

1.2 Project Objectives

The main objective of the **IaC Provisioning for Finance System** project is to automate the process of infrastructure deployment and management for financial applications using modern DevOps practices.

The key objectives of the project are as follows:

- **To automate infrastructure provisioning using Infrastructure as Code (IaC).** The project aims to replace manual infrastructure setup with automated scripts that define and create infrastructure resources efficiently.
- **To create a reliable and repeatable deployment process.**
  By using predefined configuration scripts, the same infrastructure can be deployed consistently across development, testing, and production environments.
- **To reduce manual configuration errors.**
  Automation helps minimize human errors that may occur during manual infrastructure configuration.
- **To enable scalable infrastructure for financial applications.**
  The system is designed to support scalable infrastructure that can handle increasing workloads and system requirements.

1.3 Scope of the Project

The scope of this project is to design and implement an automated infrastructure provisioning system for financial applications using **Infrastructure as Code (IaC)**.

The project focuses on automating the creation and management of infrastructure resources to ensure consistency, scalability, and reliability. It includes the use of **IaC tools, containerization technologies, and version control systems** to manage infrastructure and application deployment efficiently.

The system allows developers to define infrastructure configurations through code, which can then be automatically executed to create the required environment. This approach reduces manual effort, improves deployment speed, and ensures consistent infrastructure setup across different environments.

However, the project mainly focuses on **infrastructure provisioning and deployment automation** rather than building a full-scale financial application system.

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2. Proposed Solution

2.1 Key Features

The proposed system includes the following key features:

- Automated infrastructure provisioning
- Containerized application deployment
- Version-controlled infrastructure configuration
- CI/CD pipeline automation
- Scalable infrastructure architecture
- Reduced deployment errors
- Faster application deployment

2.2 Overall Architecture / Workflow

The system follows an automated workflow to provision and deploy infrastructure for financial applications

1. **CodeDevelopment**
   Developers write application code and infrastructure configuration scripts.
2. **Version Control**
   The project files are stored and managed using **GitHub** for collaboration and version tracking.
3. **CI/CD Pipeline Trigger**
   Whenever code changes are pushed to the repository, the automated pipeline gets triggered.
4. **Infrastructure Provisioning**
   Infrastructure resources such as servers and networking components are created automatically using **Infrastructure as Code scripts**.
5. **Containerization**
   The application is packaged into **Docker containers**, ensuring consistent execution across environments.
6. **Deployment**
   The containerized application is deployed to the provisioned infrastructure.
7. **Monitoring and Maintenance**
   The deployed system can be monitored and maintained to ensure proper functionality and performance.

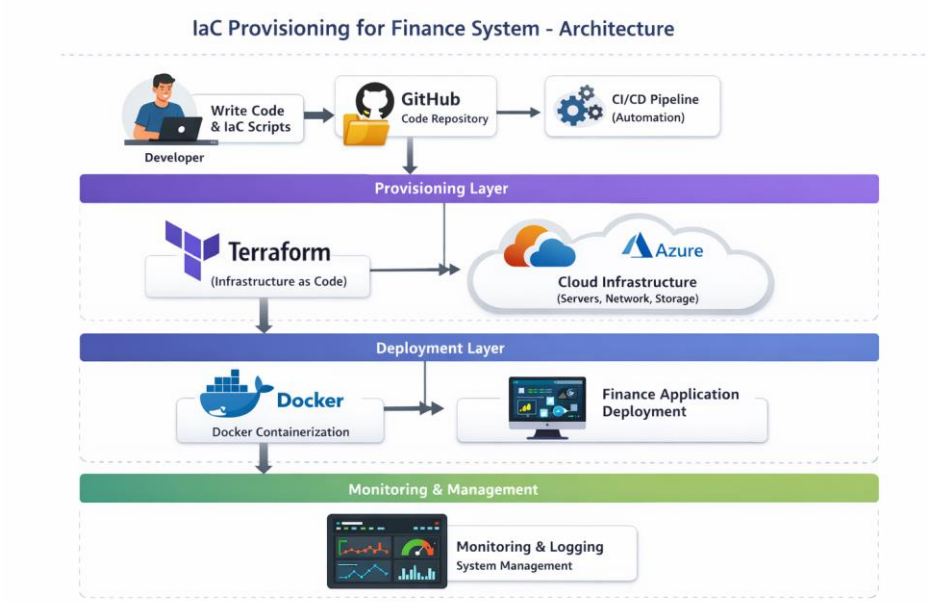

Fig 2.2 Over all Architecture

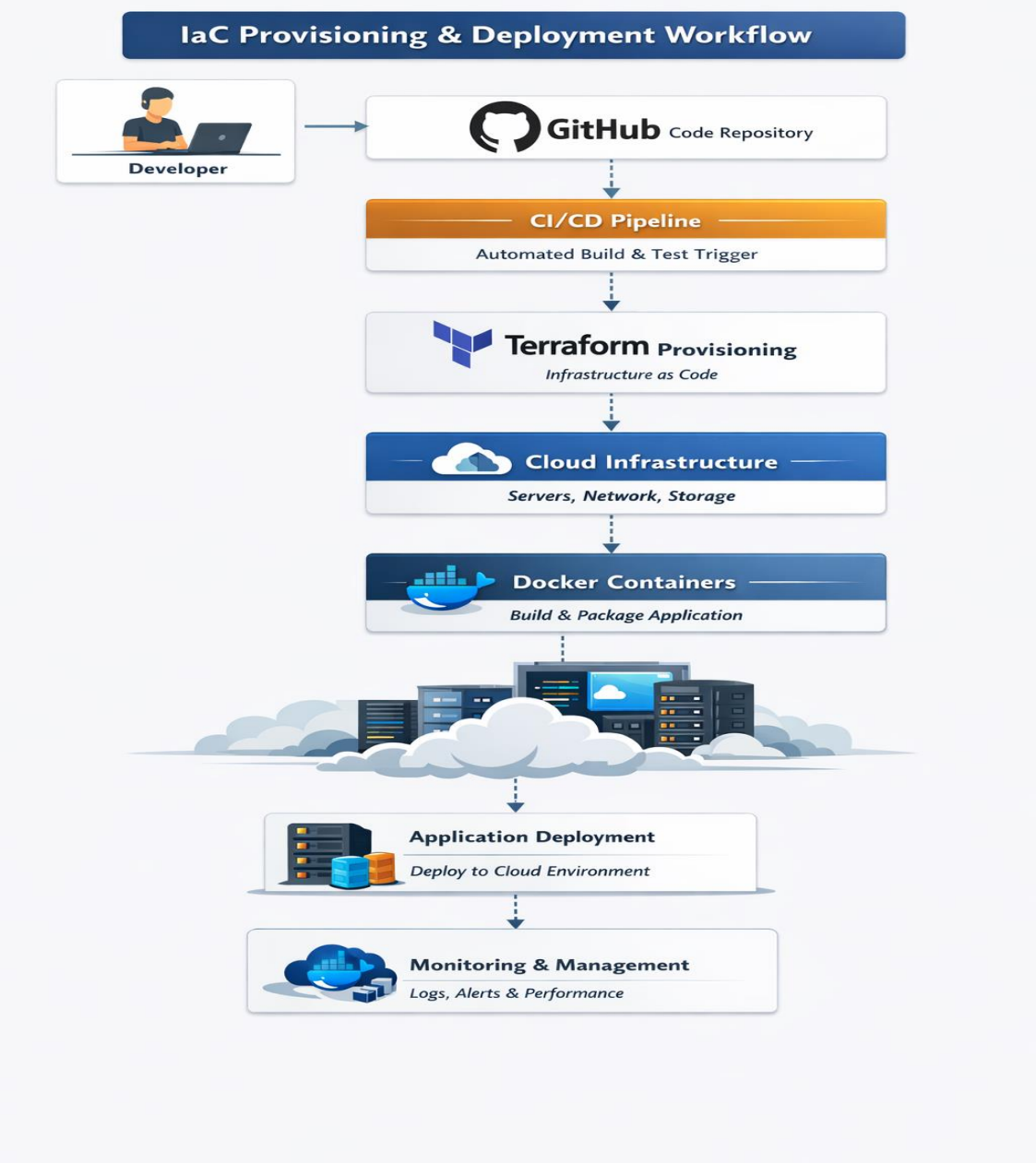

Fig.2.2 Workflow Diagram

This workflow ensures **automation, reliability, and scalability in infrastructure management**.

2.3 Tools & Technologies Used

Tool / Technology	Purpose
Terraform	Infrastructure provisioning using IaC
Docker	Containerization of application
Git	Version control
GitHub	Code repository and collaboration
CI/CD Tools	Automated build and deployment
Linux	Server operating system
Cloud Environment	Hosting and running infrastructure

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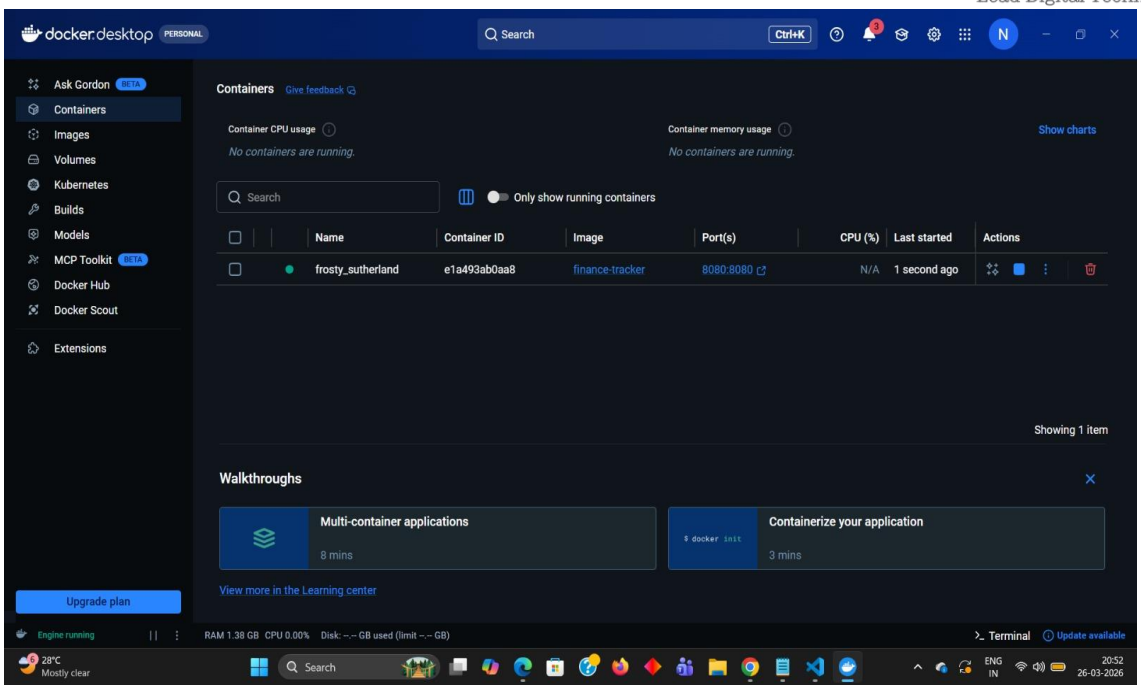
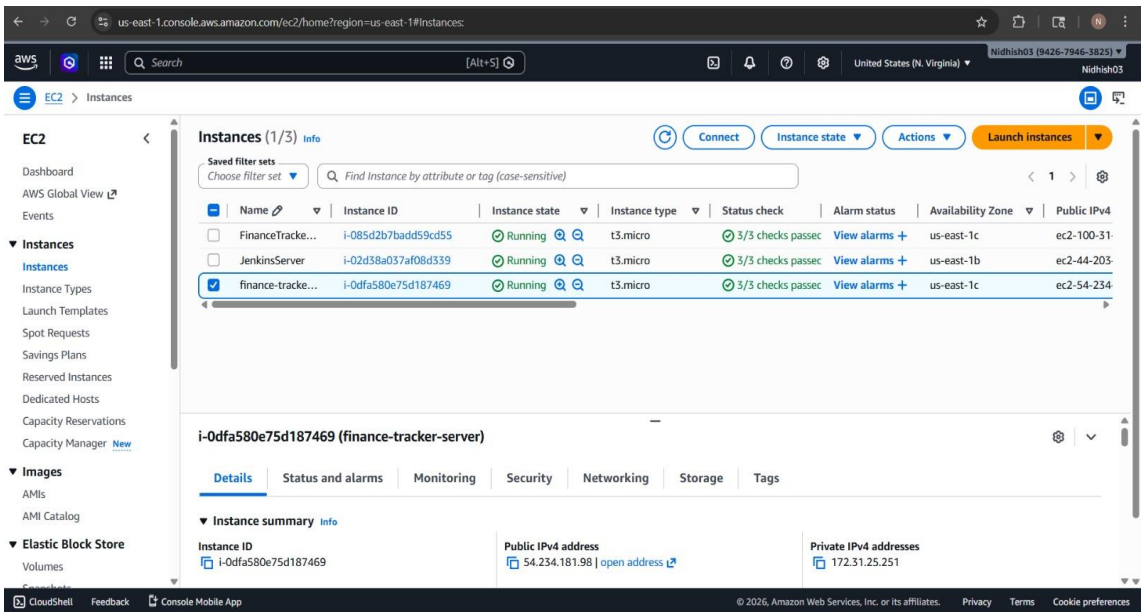
3. Results & Output

3.1 Screenshots / Outputs

The implementation of the project produced several outputs demonstrating the successful automation of infrastructure provisioning. These outputs include:

- GitHub repository containing project source code and configuration files.
- Execution results of Infrastructure as Code scripts for provisioning infrastructure.
- Docker container build and deployment outputs.
- Running application interface after deployment.


docker desktop
Search
Ask Gordon
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit
Docker Hub
Docker Scout
Extensions
Containers
Container CPU usage
No containers are running.
Container memory usage
No containers are running.
Show charts
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
frosty_sutherland
e1a493ab0aa8
finance-tracker
8080:8080
N/A
1 second ago
Showing 1 item
Walkthroughs
Multi-container applications
8 mins
Containerize your application
3 mins
View more in the Learning center
Upgrade plan
Terminal
Update available


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
United States (N. Virginia)
Nidhish03
EC2
Instances
Dashboard
AWS Global View
Events
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Instances (1/3)
Connect
Instance state
Actions
Launch instances
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
FinanceTracke...
i-085d2b7badd59cd55
Running
t3.micro
3/3 checks passed
View alarms
us-east-1c
JenkinsServer
i-02d38a037af08d339
Running
t3.micro
3/3 checks passed
View alarms
us-east-1b
finance-tracke...
i-0dfa580e75d187469
Running
t3.micro
3/3 checks passed
View alarms
us-east-1c
i-0dfa580e75d187469 (finance-tracker-server)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Instance ID
i-0dfa580e75d187469
Public IPv4 address
54.234.181.98 | open address
Private IPv4 addresses
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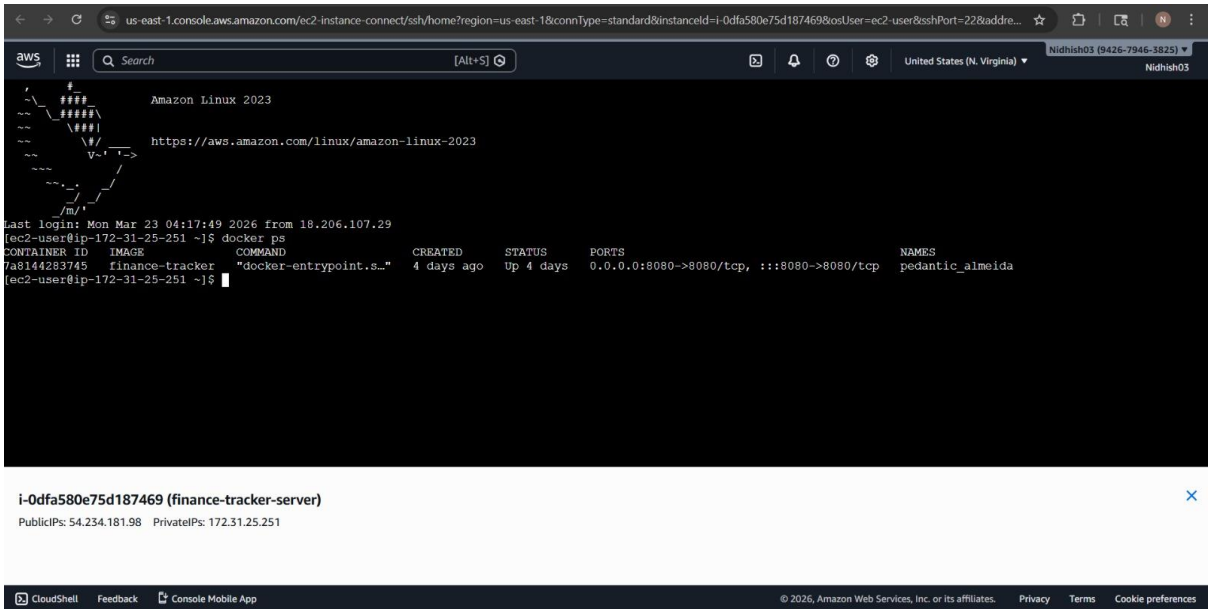
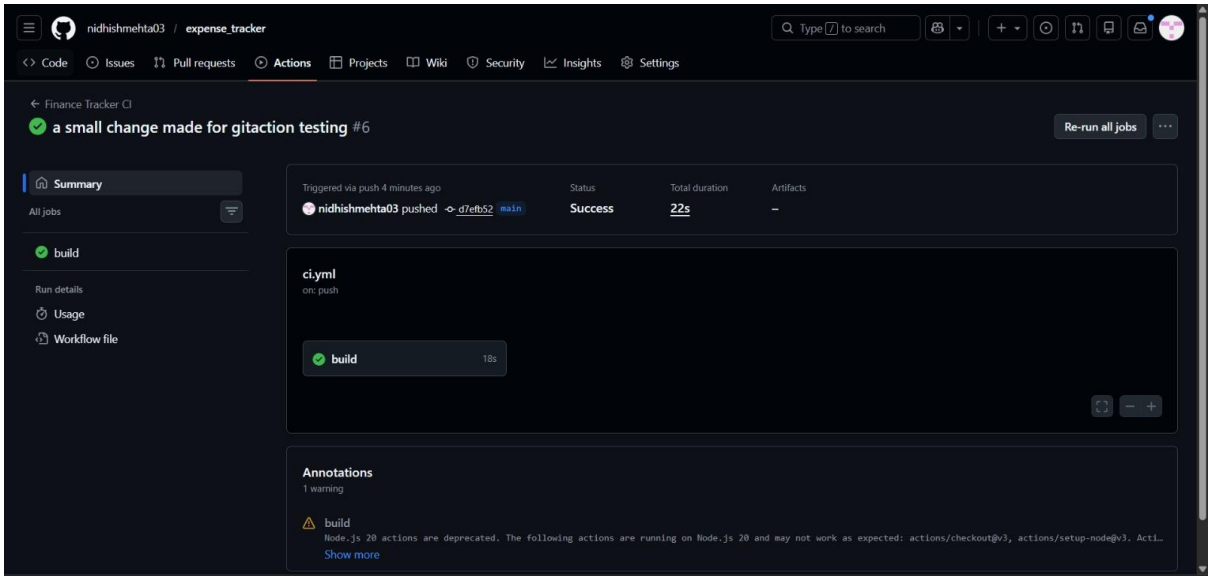
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences





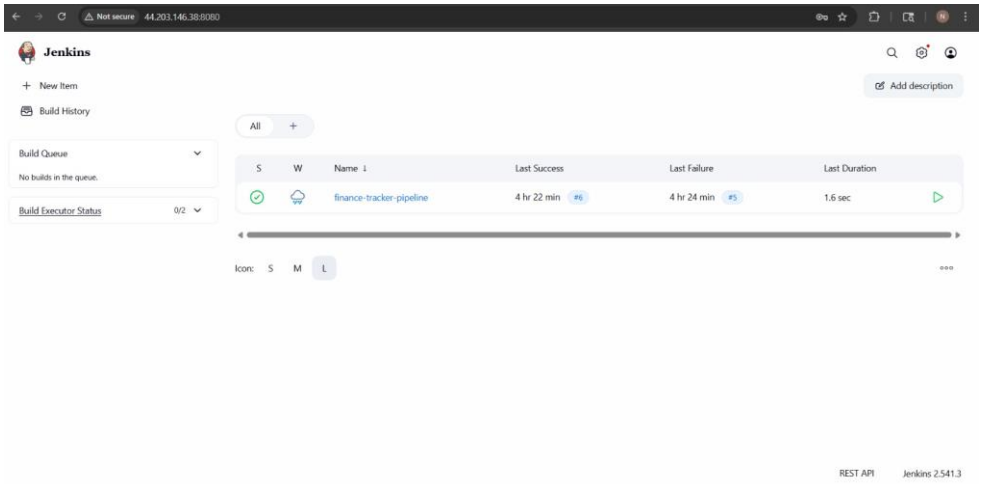
Relevant screenshots such as **terminal outputs, Docker build results, GitHub repository view, and application execution screens** can be included in this section.

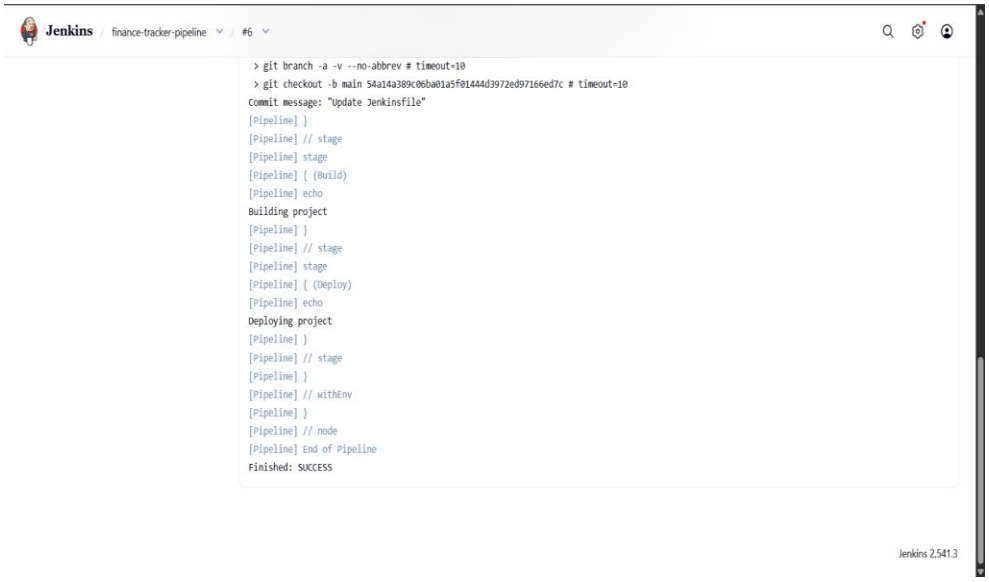
3.2 Reports / Dashboards / Models

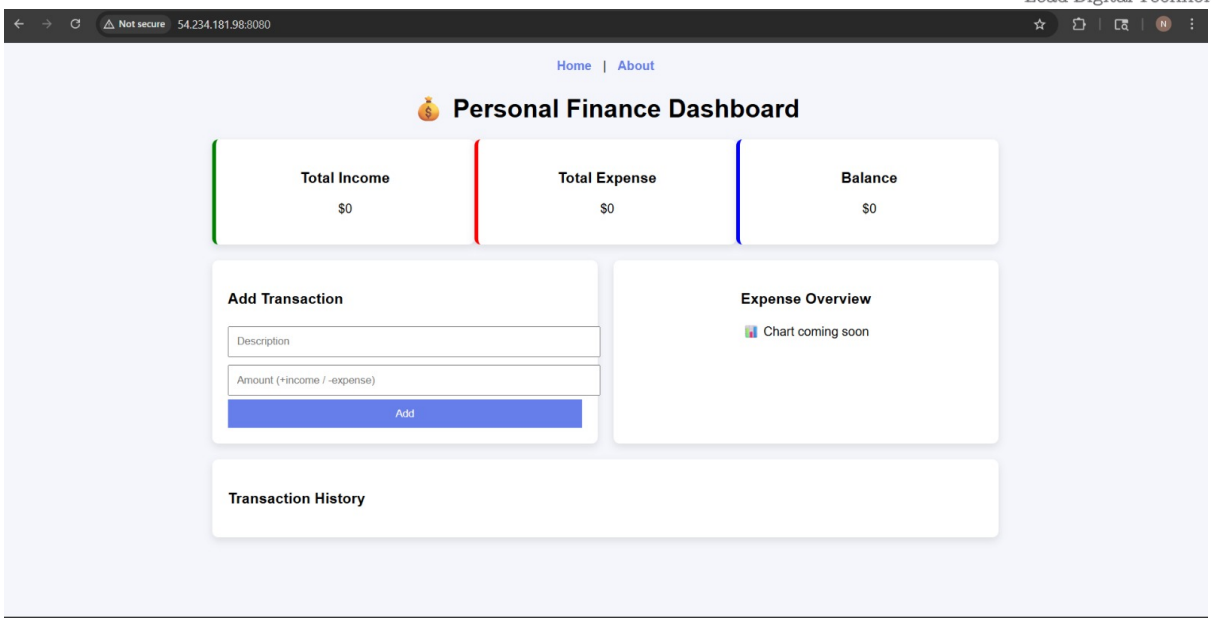
The project also generates various reports and models that help in understanding the system functionality. These include:

- CI/CD pipeline execution reports.
- Infrastructure provisioning logs.
- System architecture diagrams.
- Workflow and UML diagrams.

These models help visualize the complete infrastructure provisioning process and system in







3.3 Key Outcomes

The implementation of the **IaC Provisioning for Finance System** project resulted in several important outcomes that improved the overall infrastructure management process.

- **Successful automation of infrastructure provisioning**
  The project successfully implemented Infrastructure as Code (IaC) to automate the provisioning of infrastructure resources. This allowed infrastructure components such as servers and configurations to be created automatically through predefined scripts.
- **Faster deployment compared to manual setup**
  Automated provisioning significantly reduced the time required to set up infrastructure. Tasks that previously required manual configuration can now be completed quickly using automated workflows.
- **Consistent configuration across different environments**
  By using IaC scripts, the same infrastructure configuration can be deployed across development, testing, and production environments. This ensures consistency and reduces environment-related issues.
- **Reduced chances of configuration errors**
  Automation minimizes the risk of human errors that may occur during manual infrastructure setup. Predefined configuration scripts ensure that the infrastructure is deployed according to standard specifications.

- **Improved efficiency in managing financial system infrastructure**
  The system improves overall efficiency by simplifying infrastructure management, reducing manual effort, and enabling faster deployment of financial applications.

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4. Conclusion

The **IaC Provisioning for Finance System** project successfully demonstrates how Infrastructure as Code can be used to automate the provisioning and management of infrastructure for financial applications. The project focuses on replacing traditional manual infrastructure setup with an automated approach using modern DevOps tools and practices.

By integrating technologies such as **Terraform, Docker, Git, and CI/CD pipelines**, the system enables automated infrastructure deployment, containerized application management, and efficient configuration control. This approach ensures consistent infrastructure environments, reduces manual configuration errors, and speeds up the deployment process.

The implementation of Infrastructure as Code also improves scalability and flexibility, allowing infrastructure resources to be managed more efficiently. Overall, the project highlights the importance of **automation, DevOps practices, and infrastructure management techniques** in building reliable and scalable financial systems.

Through this project, valuable knowledge and practical experience were gained in **cloud infrastructure provisioning, containerization, version control, and automated deployment workflows**, which are essential skills in modern software development and DevOps environments.

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5. Future Scope & Enhancements

The project can be further enhanced by implementing additional features such as:

- Integration with **Kubernetes** for container orchestration.
- Implementation of **automated monitoring tools** like Prometheus and Grafana.

- Adding **security monitoring and vulnerability scanning**.
- Deploying the system on multiple cloud platforms.

- Implementing **auto-scaling infrastructure** for handling high workloads.

These improvements can make the system more efficient, scalable, and suitable for real-world financial applications.